

★ WEEK 2 · PART 3 · PRACTICE

Cointegration, Worked by Hand

Catch a Spurious Regression, Then Find the Real Long-Run Link

The Granger-Newbold warning, and the Engle-Granger residual test

Part 2's finale, with the arithmetic shown. We expose a spurious regression - a high R-squared betrayed by a near-zero Durbin-Watson - then use the residual to tell it apart from genuine cointegration, where a stationary combination reveals a real, tradeable equilibrium.

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PART 3 · APPLICATIONS

Fake Link, or Real Equilibrium?

Case A: two independent random walks produce a high R-squared and a huge t-statistic - a spurious regression the Durbin-Watson exposes. Case B: when the link is genuine, the residual is stationary - cointegration - and the Engle-Granger test confirms it. The residual is the tell. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

Regressing random walks is a trap

Two unrelated non-stationary series can show a high R-squared and a huge t. The usual asymptotics simply fail.

2

The residual is the tell

A near-zero Durbin-Watson - a non-stationary residual - flags a spurious regression. Granger-Newbold: suspect it when R-squared exceeds DW.

3

Cointegration is the real thing

When a stationary combination exists, the levels genuinely belong together. Engle-Granger tests it by checking the residual.

Everything here is a calculator exercise: one spurious warning, one residual test.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a regression of two $I(1)$ series.
- Read the residual - DW, then ADF.
- Then judge spurious or genuine.

What you should watch

- How R-squared exceeds DW in the trap.
- How the residual reveals the truth.
- How a stationary residual means cointegration.

The Spurious Trap

A great fit that means nothing

Regress one random walk on another, entirely unrelated one, and the output looks stellar: a high R-squared and a t-statistic that screams significance. But it's an illusion. The Durbin-Watson statistic, near zero, exposes it - the residuals are themselves a wandering random walk. We compute the warning and see the trap.

The regression output

- R-squared of 0.85.
- A t-statistic of 18.
- But a Durbin-Watson of just 0.15.

What we'll do by hand

- Apply the Granger-Newbold rule.
- Turn the DW into a residual autocorrelation.
- See why the fit is fake.

The Spurious Symptoms

Impressive numbers, no substance

two independent random walks $\rightarrow$ high R^2 , huge t – but the relationship is **spurious**

Why it happens

- Both series trend, so they seem to co-move.
- OLS reads shared drift as a relationship.
- The standard errors are badly wrong.

The giveaway

- The residuals don't settle - they wander.
- So the Durbin-Watson collapses toward zero.
- That's the signal to distrust the fit.

The Granger-Newbold Rule

When R-squared beats Durbin-Watson

Granger–Newbold: $R^2 = 0.85 > DW = 0.15 \Rightarrow$ suspect a spurious regression

The rule

- Compare R-squared to the Durbin-Watson.
- Here, 0.85 against 0.15.
- R-squared far exceeds DW.

The verdict

- That inequality is the classic red flag.
- Suspect a spurious regression.
- The high R-squared is not to be trusted.

What the DW Reveals

A residual near a unit root

$$DW \approx 2(1 - \hat{\rho}) \Rightarrow \hat{\rho} = 1 - \frac{0.15}{2} = \mathbf{0.925} \text{ (residuals } \approx \text{ a unit root)}$$

The conversion

- Durbin-Watson is about two times one minus rho.
- So rho is one minus DW over two.
- Here, 0.925.

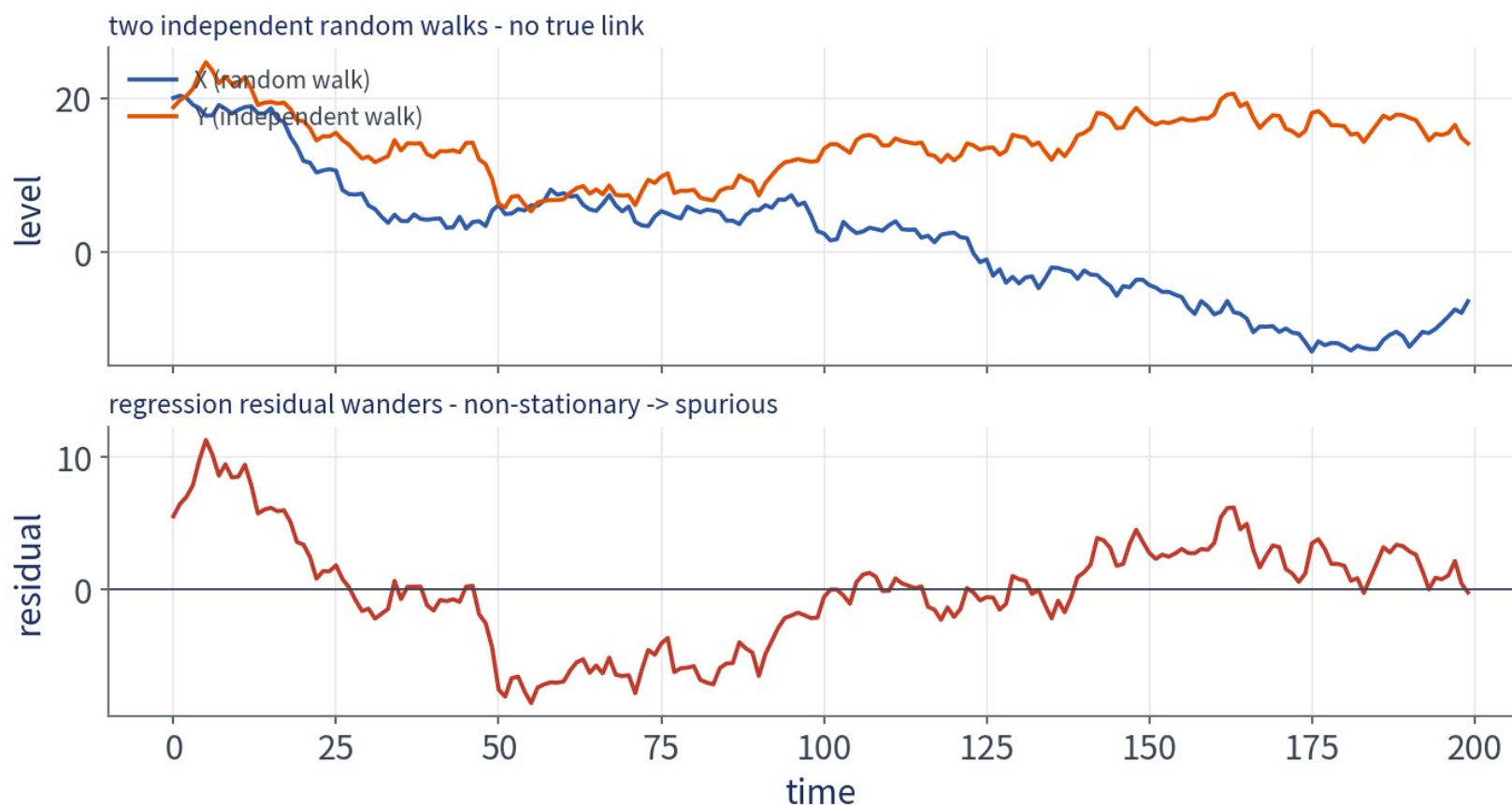
The meaning

- The residuals are almost perfectly persistent.
- That's essentially a unit root.
- Non-stationary residuals mean a fake link.

A Fit Built on Sand

The top panel shows two independent random walks that drift and appear to track each other - enough to give a high R-squared. The bottom panel shows the regression residual, which wanders without ever returning to zero. A non-stationary residual is the signature of a spurious regression: the relationship

Case A: high R-squared, but the residual never mean-reverts



Four Habits That Keep You Honest

Avoiding the trap

The discipline

- Check each series for a unit root first.
- Inspect the residual, not just the R-squared.
- Watch the Durbin-Watson.

The escape

- Difference to returns when there's no real link.
- But differencing throws away the long run.
- If the levels truly belong together, don't.

The Genuine Link

When the residual is stationary

Sometimes two non-stationary series really do move together - a stock and its hedge, two related futures. Then their combination, the spread, is stationary even though each series wanders. That's cointegration. The Engle-Granger test checks it by testing the residual for a unit root. We run it and find a real, tradeable

The genuine pair

- Two $I(1)$ series driven by a common trend.
- Regressing gives a slope of 1.05.
- The residual is the spread.

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What we'll do by hand

- Test the residual for stationarity.
- Compare to the Engle-Granger cutoff.
- Confirm cointegration.

Cointegration, Formally

Two wanderers, one stationary combination

$y_t, x_t \sim I(1)$ but $e_t = y_t - \beta x_t \sim I(0) \Rightarrow$ **cointegrated** (a stationary combination)

The definition

- Each series is integrated of order one.
- Yet some combination is stationary.
- That combination is the equilibrium.

Why it's special

- The levels share a long-run anchor.
- Differencing would discard exactly that.
- Cointegration keeps it.

The Engle-Granger Test

Test the residual for a unit root

$$\text{ADF on } \hat{e}_t : t = -4.20 < \underbrace{-3.34}_{\text{EG 5\%}} \Rightarrow \text{residual stationary} \Rightarrow \text{cointegrated}$$

The two steps

- First, regress one series on the other.
- Then run an ADF on the residual.
- A stationary residual means cointegration.

The special cutoff

- The residual is estimated, not observed.
- So the cutoff is -3.34, not -2.86.
- At -4.20, we clear it - cointegrated.

Spurious vs Cointegrated

The residual decides

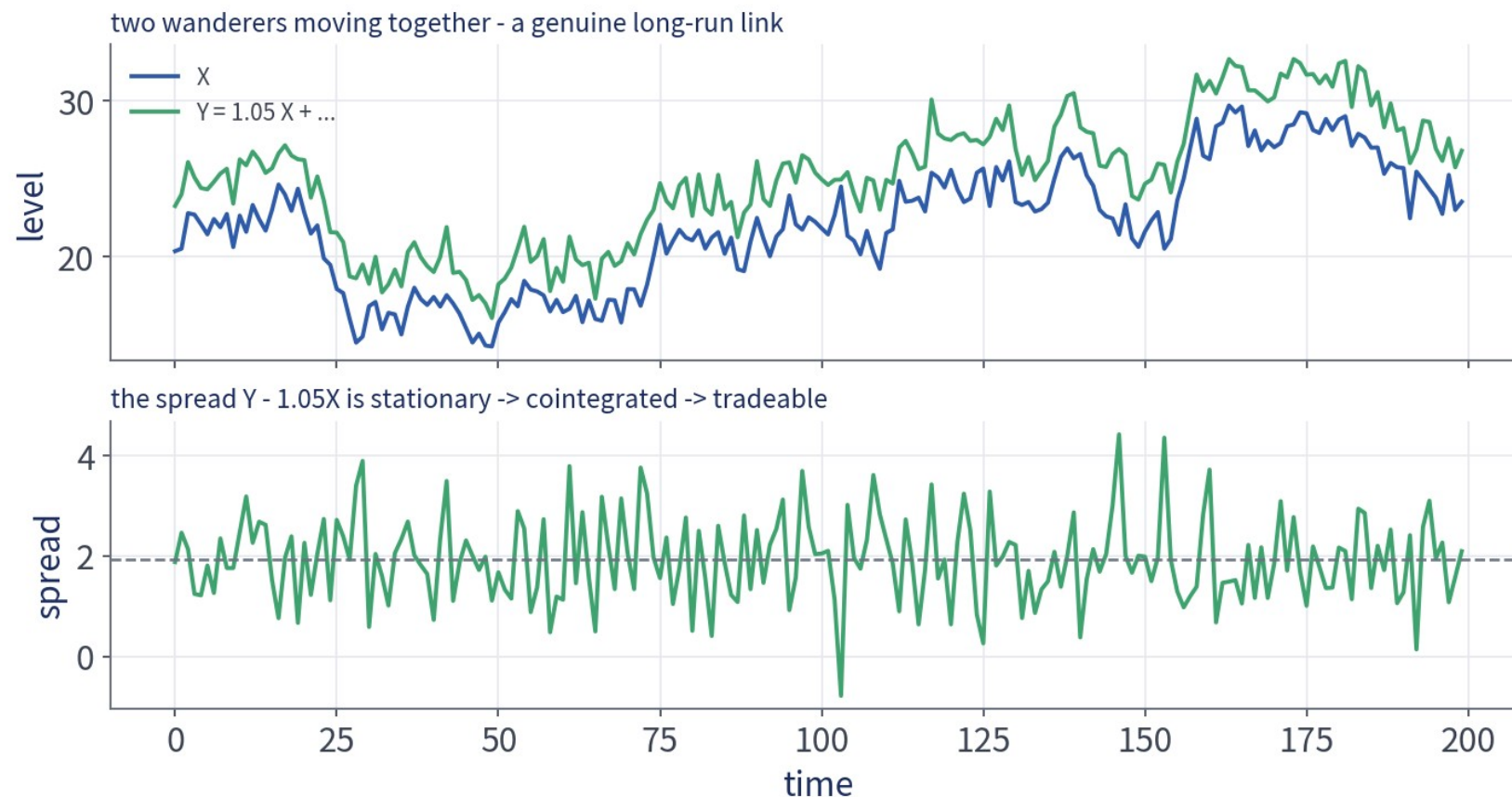
	spurious (Case A)	cointegrated (Case B)
Durbin-Watson	0.15 (near 0)	1.85 (healthy)
residual ADF	fails (~ -1.1)	rejects (-4.20)
residual is	non-stationary	stationary
the link is	fake	real, tradeable

Same-looking regressions, opposite truths - and the residual tells them apart. A wandering, non-stationary residual is spurious; a stationary, mean-reverting one is cointegration.

A Spread That Comes Home

The top panel shows two wandering series held together by a common trend - a genuine long-run link. The bottom panel shows their spread, which oscillates tightly around its mean instead of drifting. That stationary residual is cointegration, and its mean reversion is exactly what a pairs trade exploits.

Case B: the residual mean-reverts - a real equilibrium



From Equilibrium to a Trade

ECM and pairs trading

The error correction model

- When the spread strays, it gets pulled back.
- The ECM measures how fast.
- That pull is the tradeable signal.

Pairs trading

- Enter when the spread is far from its mean.
- Exit as it reverts.
- Cointegration is what makes it work.

The Cointegration Workflow

Suspect, test, then trade
The routine protects you and then pays you. When regressing non-stationary series, suspect a spurious result - check the Durbin-Watson and apply Granger-Newbold. Then test the residual properly: an Engle-Granger ADF that rejects means the residual is stationary and the series are cointegrated. A stationary spread is a real equilibrium - and mean reversion in that spread is what a pairs

The trap (Case A)

- R-squared 0.85 beats DW 0.15.
- Residual autocorrelation 0.925.
- Spurious - the link is fake.

Si The link (Case B)

- Residual ADF -4.20 clears -3.34.
- Stationary spread, DW 1.85.
- Cointegrated - and tradeable.

Reproduce the Trap and the Link

Direct Claude Code to check your arithmetic

GOAL Expose a spurious regression, then confirm a cointegrated pair.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - spurious:

- * simulate two independent random walks, regress one on the other; print R-squared, t, and DW;
- * apply the R-squared > DW rule.

CASE B - cointegration:

- * simulate a cointegrated pair (common trend);
- * run the Engle-Granger test (regress, then ADF the residual) and compare to -3.34.

Then plot both pairs with their residual/spread.

Check against the slides

- spurious: $R^2 0.85 > DW 0.15$.
- $\rho \sim 0.925$ (near unit root).
- cointegrated: $ADF -4.20 < -3.34$.
- spread is stationary.

Deliverable: `part3_practice.py` whose printed numbers match the spurious warning and the EG test.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What flags a spurious regression?

**A high R-squared
with a near-zero
Durbin-Watson**

Q2 What does cointegration require?

**A stationary
combination of non-
stationary series**

Q3 How does Engle-Granger confirm it?

**An ADF on the
residual that rejects
a unit root**

All three come from the Granger-Newbold rule, the definition of cointegration, and the residual test.

Three Pitfalls to Avoid

Where cointegration trips people

Common mistakes

- "A high R-squared means a real relationship."
- "Always difference to be safe."
- "Use standard ADF cutoffs on the residual."

The reality

- Check the residual - a low DW means spurious.
- Differencing discards a genuine long-run link.
- The residual is estimated - use -3.34, not -2.86.

Two Calculations to Remember

Part 3 applications in one screen

1

A high R-squared can be a lie

Two independent random walks gave an R-squared of 0.85 and a t of 18 - but a Durbin-Watson of 0.15. Since R-squared exceeds DW, Granger-Newbold flags it spurious; the implied residual autocorrelation of 0.925 confirms a near-unit-root residual.

2

A stationary residual is the real thing

For a genuine pair, the Engle-Granger ADF on the residual was -4.20, clearing the special -3.34 cutoff - so the residual is stationary and the series cointegrated. The Durbin-Watson of 1.85 agrees. The spread is a real equilibrium.

3

The residual is always the tell

Same-looking regressions, opposite truths: a wandering residual is spurious, a mean-reverting one is cointegration. That stationary spread is what the error correction model and pairs trading turn into a strategy.

Week 2 complete — from stationarity through unit roots to cointegration, the foundation every later model rests on.